IMDb Sentiment Analysis

Sanidhya Kaushik
Indian Institute of Science
sanidhyak@iisc.ac.in

Agam Ghotiya
Indian Institute of Science
agamghotiya@iisc.ac.in

Abstract

This study evaluates classical TF-IDF based models against recurrent neural networks (RNN/LSTM) for sentiment classification using the IMDb dataset. To optimize recurrent architectures, we utilize a Global Max Pooling strategy to mitigate signal dilution caused by sequence padding. Empirical results demonstrate that a tuned Linear SVM and Logistic Regression achieve the highest accuracy (88.0%), outperforming neural models. Qualitative analysis via Latent Semantic Analysis (LSA) further illuminates the feature distribution of the textual data[1].

1 Introduction

Sentiment analysis of long form text requires models to extract meaningful signals from noisy data. While deep learning is often preferred for NLP, high dimensional linear models remain highly competitive on specific datasets. We present a modular pipeline to benchmark these approaches and analyze the impact of architectural pooling on recurrent networks.

2 Methodology

2.1 Dataset and Preprocessing

Text was normalized via lowercasing and removal of punctuation/HTML tags. For deep learning, we utilized a vocabulary of 10,000 tokens, with sequences truncated or padded to a fixed length of 250. The IMDb dataset consists of 50,000 reviews with a balanced binary sentiment distribution. Exploratory analysis showed that both positive and negative reviews maintain similar distributions in length (see figure 1), averaging approximately 230 words per sample. The average word length for both positive and negative review also follow the same distribution (see figure 2).

2.2 Models

- **Classical:** TF-IDF vectorization (max 5,000–10,000 features) was paired with Multinomial Naive Bayes, Logistic Regression, and Linear SVM. Hyperparameters were optimized via 5-fold grid search.
- **Neural:** We implemented a vanilla RNN and an LSTM. To improve upon standard many to one configurations, a **Global Max Pooling** layer was added after the recurrent layers to capture the most significant hidden states across the temporal dimension.

[1]The code is available here

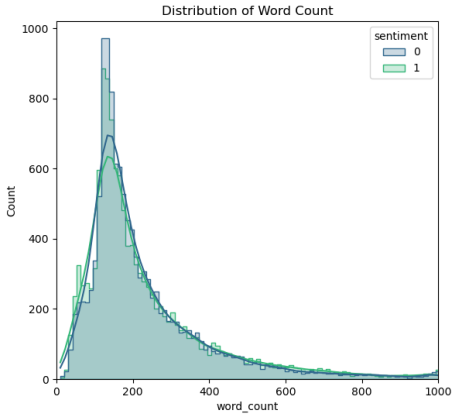

Figure 1: Histogram of word Counts

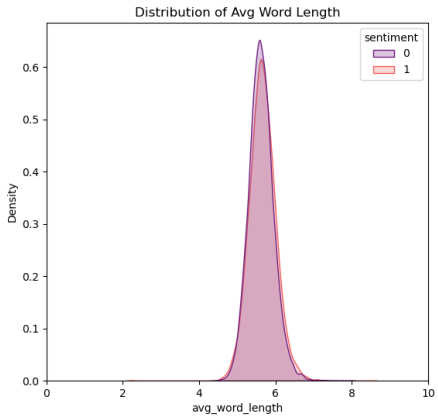

Figure 2: Histogram of average word length

3 Results and Analysis

3.1 Qualitative Results

To characterize the feature space, we performed a qualitative analysis using TF-IDF weighted word clouds and Latent Semantic Analysis (LSA). As shown in Figure 3, 4 and 5, discriminative tokens such as 'excellent' and 'waste' drive class separation. However, the 2D LSA projection (Figure 6) reveals significant spatial overlap between classes, suggesting that a soft margin approach is necessary for linear classifiers.

Figure 3: Word cloud for negative reviews

Figure 4: Word cloud for positive reviews

3.2 Quantitative Performance

Classical models consistently outperformed neural models. Figure 7 - 14 show the corresponding

Model	Accuracy	Precision	Recall	F1-Score
SVM (Tuned)	87.9%	0.8682	0.8937	0.8808
Logistic Regression (Tuned)	88%	0.8664	0.8983	0.8821
Naive Bayes (Tuned)	84.9%	0.8526	0.844	0.8483
SVM	85.79%	0.8594	0.8558	0.8576
Logistic Regression	87.65%	0.8729	0.8814	0.8771
Naive Bayes	83.92%	0.8519	0.8142	0.8326
RNN	81.23%	0.8523	0.7556	0.8010
LSTM	85.23%	0.8556	0.8331	0.8442

Table 1: Quantitative Results

confusion matrices

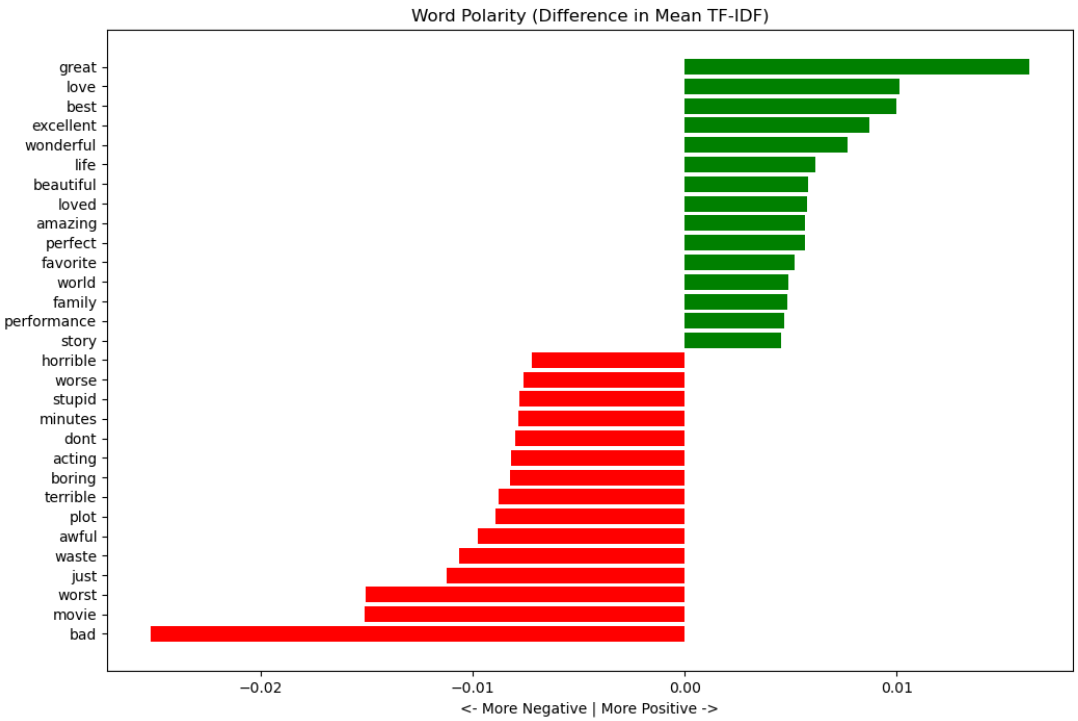

Figure 5: Tokens having highest polarity

3.3 Why Classical Models Outperform Neural Models?

The following are the potential reasons why classical models outperform neural models

- **Lack of Pre-training:** The neural models here are largely overparametrized and haven't been pre-trained. The feature to signal ratio is high as well.
- **Syntax vs Meaning:** The TF-IDF features largely ignore the order and focus on the lexical part of the reviews whereas neural models are designed to be sensitive to flow of the language.
- **Information Bottleneck:** Recurrent architectures (especially RNNs) suffer from the issues of information bottleneck which significantly reduces their performance.

4 Discussion

We would like to address the following two questions in this section

- **RNN and Global Max Pooling:** While training vanilla RNN, we observed that it was unable to propagate gradients properly and thus the accuracy remained largely around 50%. Thus we had to necessarily employ global max pooling to facilitate learning. On the other hand the jump in performance of LSTM upon using global max pooling was approximately 2-3%.
- **Different Kernel Choices for SVMs:** In this study we only used Linear SVMs and didn't experiment with other kernel choices like RBF due to the following two reasons:
  - **Cover's Theorem:** It states that even though the 2D scatter plot shows high inter class overlap, the samples will be linearly separable in a high dimensional space (5000 dimensional here) with a very high probability.
  - **Computational Overhead:** Computing the RBF kernel matrix is a O(N^2) operation but LinearSVC is optimized using LibLinear library which scales linearly.

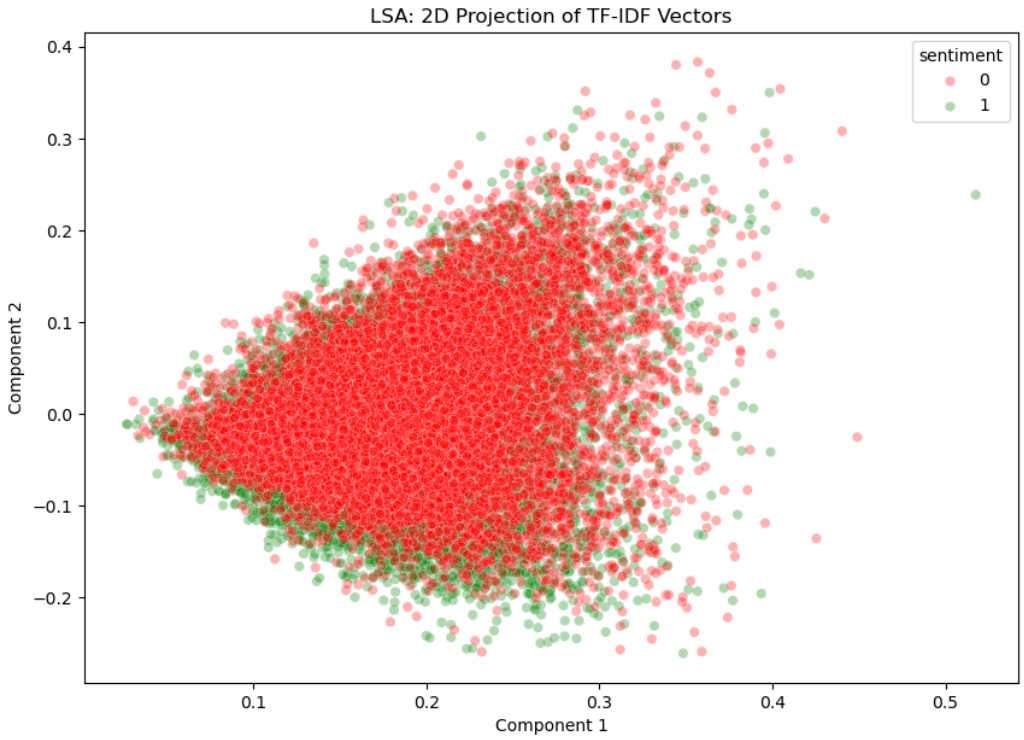

Figure 6: 2D scatter plot of reviews

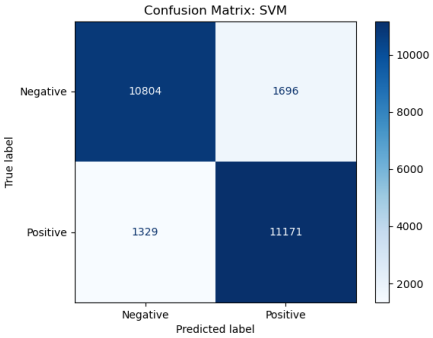

Figure 7: Confusion matrix of tuned SVM

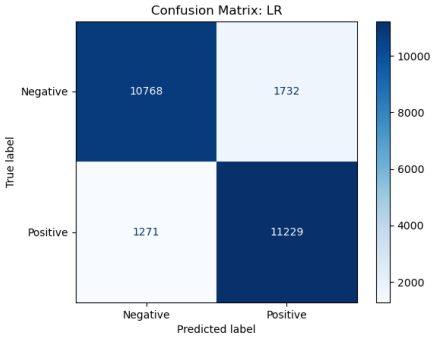

Figure 8: Confusion matrix of tuned logistic regressor

5 Conclusion

In this study, we conducted a systematic benchmarking of classical machine learning and deep recurrent architectures for sentiment classification on the IMDb dataset. Our empirical results demonstrate that high dimensional linear models, specifically the tuned Linear SVM and tuned Logistic Regressor, provide a superior and more robust solution for this task, achieving a peak accuracy of 88.0%. The performance gap between the classical and neural models suggests that for long form text of this nature, global lexical importance represented by sparse TF-IDF features is a more reliable indicator of sentiment than the local sequential dependencies captured by RNNs and LSTMs trained from scratch. We hypothesize that the neural models were limited by a high parameter to sample ratio, leading to overfitting on noise, whereas the convex optimization of the Linear SVM effectively identified an optimal separating hyperplane.

Architecturally, we demonstrated that the integration of a Global Max Pooling layer is essential for recurrent models in this domain. By capturing the most salient hidden states across the temporal

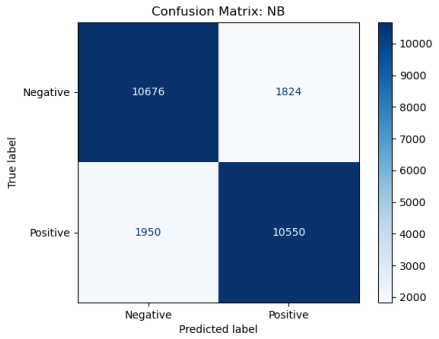

Figure 9: Confusion matrix of tuned Naive Bayes

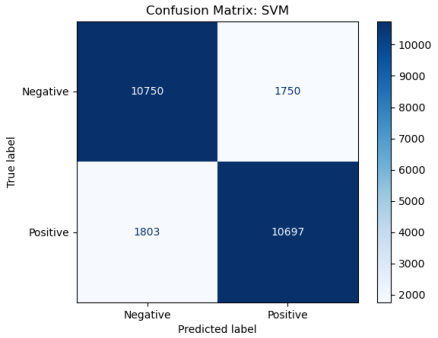

Figure 10: Confusion matrix of SVM

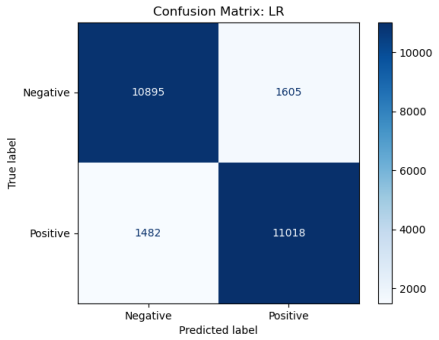

Figure 11: Confusion matrix of Logistic Regressor

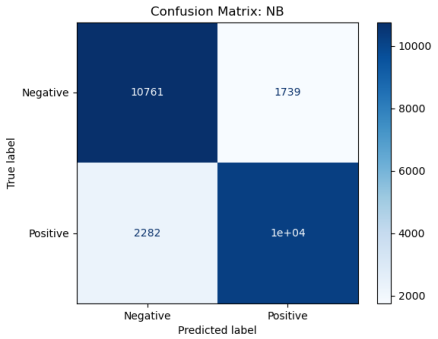

Figure 12: Confusion matrix of Naive Bayes

dimension, this strategy successfully mitigated the signal dilution typically caused by sequence padding.

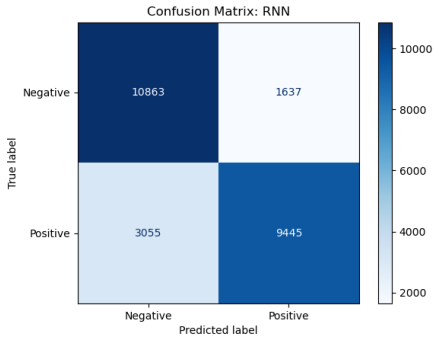

Figure 13: Confusion matrix of RNN

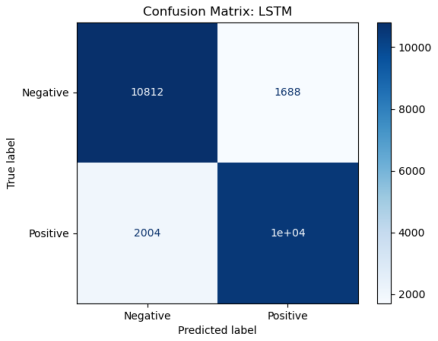

Figure 14: Confusion matrix of LSTM